

3D Reconstruction of a Soft Object Surface and Contact Areas in Hand-Object Interactions

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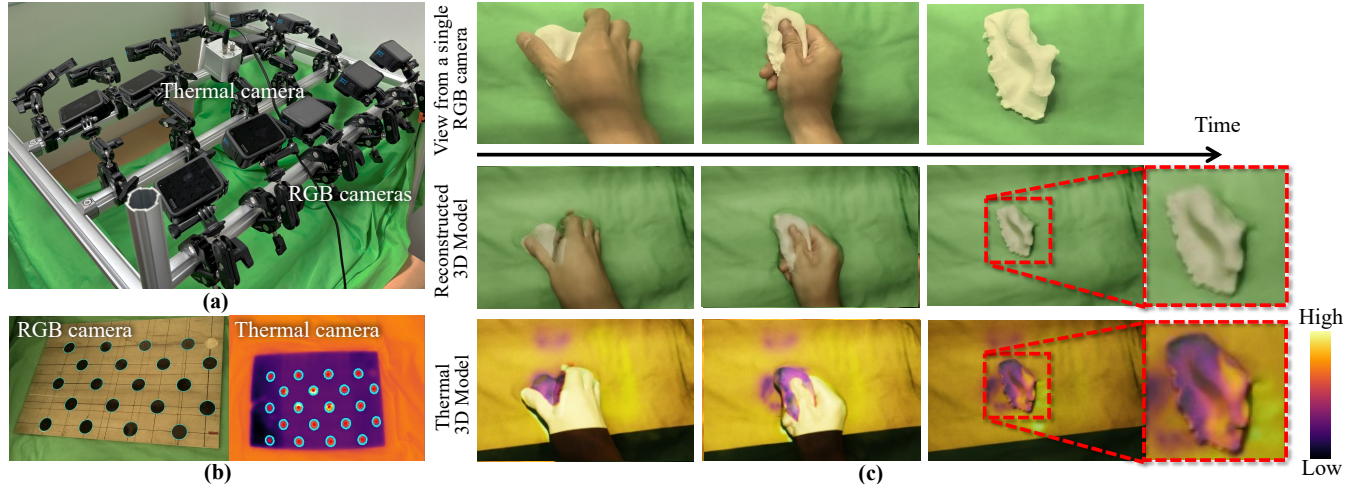


Figure 1: (a) Our setup consists of 14 RGB cameras and a thermal camera looking down from above. Both types of cameras detect the black-painted aluminum disks for calibration. (c) The results of our proposed method. With time passing left to right, the first row is hand-object interaction from one of the RGB camera views, the second shows reconstructed 3D meshes with an RGB texture, and the third shows those with a thermal texture.

Abstract

In Hand-Object Interactions (HOIs), contact information between the hand and the object can be utilized to estimate the area of deformation area and to improve the relative position of the hand and the object. However, mapping contact information to unknown and deformable objects is still challenging due to their shape transformation. In this study, we present a preliminary attempt to reconstruct the surface of a soft object and identify the contact area on that surface in HOIs. By using our system composed of multiple RGB cameras and a single thermal camera, our proposed method successfully visualizes the areas on the object's surface.

CCS Concepts

• Computing methodologies → Shape modeling.

Keywords

Hand-Object Interactions, soft object, thermal camera

ACM Reference Format:

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1 Introduction

Hand-Object Interactions (HOIs) are essential not only for everyday life but also for professional skills such as art and chef's cooking. Since camera sensors do not interfere with free hand movements, high-fidelity 3D reconstruction of dynamic HOIs using these sensors has many applications, such as realistic computer graphics and the analysis of modeling skills with hands [de Charette and Manitsaris 2019]. Moreover, the reproduction of HOIs enables us to automate them with robotic hands [Qin et al. 2022], and learn them using virtual reality [Wu et al. 2020].

Especially, HOIs for soft objects, such as clay or dough, require human-level dexterity. Accurate reconstruction and reproduction of these interactions are expected to contribute to the development of robotic hands capable of modeling human-level movements and

preserving traditional craft skills. Since it is the contact between the hands and the object that causes deformation in HOIs, accurately estimating the deformation process in the contact areas is crucial. However, when using RGB cameras, which are commonly used, the areas are often occluded by the hands, making it difficult to observe. In such cases, thermal cameras can capture the contact areas by detecting the rise in temperature on the object's surface caused by the contact. Iwai et al. [Iwai et al. 2006] utilized this information to detect touches on objects' surfaces as a human interface. Even in the field of HOI research, it has been used to predict grasp poses accurately by optimizing the relative poses between the hand and the object [Grady et al. 2021]. However, these methods cannot be applied to HOIs that involve shape changes because they map the contact information onto rigid and pre-known objects and predict static grasp poses.

Therefore, we propose a method for reconstructing the object's surface and its contact areas using multiple RGB cameras and a thermal camera for HOIs with soft, unknown objects. We employ RGB cameras to reconstruct the surface model of the object and a thermal camera to capture the contact areas, respectively.

2 Methods and Implementation

We present a method for reconstructing the surface model of a soft object and its contact area in HOIs using RGB cameras and a thermal camera (Fig. 2). We used GoPro Hero11 Black as the RGB cameras and an Optris PI 450i with an O80 lens as the thermal camera. They are mounted looking down from a top-down view to observe deformation along gravity (Fig. 1(b)).

For the estimation of camera poses, first, we estimate their intrinsic parameters and lens distortion coefficients formulated by the pinhole camera model. Second, we estimate their extrinsic parameters (i.e. camera pose: rotation and translation) based on the orientation of the calibration board with a known pattern. The pattern needs to be visible to each camera for the subsequent processes to be based on the same coordinate system.

We employed a 4×5 asymmetric circle grid pattern, where the circles are made of black-painted aluminum disks and fixed at predefined positions on a plywood plane (Fig. 1(a)). Aluminum has a lower emissivity compared to plywood. This difference is captured as the contrast between the circle area and the background in a thermal image.

For the estimation of intrinsic and distortion parameters, we used 25 images of the board with different poses for each type of camera. We assumed that the parameters of the RGBs were the same since they were the same product. Next, we fixed the board position to establish the world coordinate system and input the board images from each camera view to obtain each pose.

Based on the estimated RGB camera poses, the depth maps are obtained. Then, 3D meshes of the hands and object are reconstructed from the maps by a surface reconstruction method using Total generalized variation minimization [Poliarnyi 2021].

By projective texture mapping of a thermal image based on its estimated pose onto the reconstructed 3D model, contact areas are visualized. The thermal images are colored using a perceptually linear colormap. A 1-degree Y rotation misalignment was found and manually corrected.

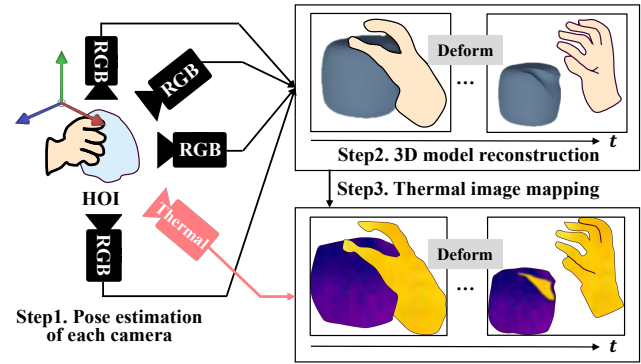


Figure 2: Overview of the proposed method

3 Experiment

We reconstructed the dynamic scene where the right hand deformed the paper clay as shown in Fig. 1(c). In Fig. 1(c), we can observe a white-yellow area that corresponds to the hand and is the hottest region in the scene. This area extends along the hand. In the close-up images, the contact areas are highlighted in yellow or orange and are enclosed by a red rectangle. The yellow areas indicate a greater heat transfer than the orange areas.

This temperature gap on the surface resulting from contact can be utilized to obtain detailed touch information. The yellow area may have been in contact for a longer period or in closer proximity, while the orange area may have had a shorter duration or slight contact. Deformation on the yellow surface is more likely to occur.

4 Conclusion and Future Work

We present a method for reconstructing the surface of a soft object without pre-scan and its contact area in HOIs. In future work, we will improve the accuracy of thermal image mapping by optimizing the pose of the thermal camera and aligning the edges of a 3D model with a thermal image. We will then quantitatively validate this method by comparing the results of painting with hands.

Acknowledgments

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